Week 12 — Final Project Report

Final report

CSE4204-8B-T04 | Northern University of Business and Technology, Khulna

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1. Introduction

Enterprise NeXus is a multi-agent AI business automation operating system for startups, small businesses, and growing teams. It brings HR, Finance, Support, Analytics, Security Intelligence, and Executive reporting into one role-based workspace.

2. Team Information

Field	Detail
Course	CSE 4204 — Mobile Computing Lab
Team	CSE4204-8B-T04
Project	Enterprise NeXus — Multi-Agent AI Business Automation OS with Security Intelligence
Supervisor	Md. Riaz Mahmud (Assistant Professor)
Institution	Northern University of Business and Technology, Khulna

Name	Student ID	Role
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3. Problem Statement

Small businesses often cannot afford separate specialists for recruitment, finance review, customer support, analytics, and executive reporting. Manual CV screening, expense checking, ticket triage, and report preparation consume time and produce inconsistent decisions. Enterprise NeXus addresses this gap with connected AI agents that work on the organisation’s own data while preserving human review and escalation.

4. Objectives

- Automate recurring department-level operations with specialised AI agents.
- Provide explainable CV scores, anomaly alerts, support responses, KPI reports, and executive briefings.
- Centralise business records in a secure, real-time workspace.
- Protect data with JWT authentication, RBAC, audit logging, rate limiting, and Row-Level Security.
- Deliver a deployable full-stack system with documentation, testing, and monitoring.

5. Methodology

Development followed the handbook workflow: idea selection, proposal, SRS, system design, UI/UX design, backend and frontend implementation, AI integration, feature completion, testing, deployment, and documentation. GitHub branches and meaningful commits supported team collaboration.

6. System Architecture

The Next.js frontend communicates with a Node.js/Express API. The API authenticates requests, applies role permissions, invokes domain agents, and stores results in Supabase PostgreSQL. Supabase Realtime delivers updates, pgvector supports semantic retrieval, Gemini provides structured generation, and n8n handles scheduled workflows and retries.

6.1 Database Design

The database uses UUID identifiers, timestamps, constraints, foreign keys, indexes, and Row-Level Security. Tables cover users, sessions, jobs, applications, transactions, orders, tickets, reviews, knowledge documents, agent tasks, and audit events.

7. AI Integration

Each specialist agent inherits a shared BaseAgent and uses scoped tools. The Executive Agent delegates cross-domain questions to HR, Finance, Support, and Analytics, then synthesises the result. Gemini native JSON output, schema validation, timeouts, retries, and explicit pending or failed states prevent fabricated results. Gemini embeddings and Supabase pgvector ground knowledge-base answers.

8. Key Features

Agent or module	Implemented capability
HR Agent	CV upload, extraction, scoring, confidence, recommendation, summary, strengths, weaknesses, and review workflow
Finance Agent	Transaction logging, import, spend analysis, anomaly severity, and explanation
Support Agent	Ticket triage, sentiment, AI reply, product-review response, escalation, resolution, and lifecycle handling
Analytics Agent	Cross-module KPI snapshots, trend analysis, and reports
Executive Agent	Natural-language queries, delegation, briefings, and executive recommendations
Security Intelligence	RBAC, audit visibility, security signals, and risk-focused monitoring

9. Testing

Testing combines automated Jest and Supertest coverage, API checks, manual UI checks across responsive layouts, AI response validation, and Postman verification. The testing evidence covers authentication, agent APIs, RAG, orchestration, graceful failure, and no-fake-data behaviour.

10. Deployment and Results

The frontend is deployed on Vercel, the backend on Render, and PostgreSQL, realtime data, and vector retrieval on Supabase. The live system demonstrates the minimum project requirements: frontend, backend, database, authentication, meaningful AI integration, deployment, testing, documentation, and GitHub workflow.

11. Challenges and Solutions

Challenge	Solution
AI output variability	Native JSON mode, schemas, normalisation, retries, and explicit failure states
Cold starts and monitoring	Dedicated health endpoints, UptimeRobot monitoring, and operational wake-up notes
Sensitive business data	JWT, RBAC, RLS, per-user scoping, validation, and audit logging
Cross-module coordination	Executive orchestration layer and n8n workflow callbacks
Support lifecycle complexity	Sentiment-aware escalation, manual resolution, and inactivity close handling

12. Future Work

- Add Legal, Marketing, and Inventory agents.

- Support additional LLM providers and local models.
- Add forecasting and deeper trend prediction.
- Introduce organisation-level multi-tenancy and richer collaboration.
- Add a native mobile client and stronger observability dashboards.

13. Conclusion

Enterprise NeXus demonstrates practical AI integration in a complete full-stack system. It connects specialised agents, secure APIs, real-time data, workflow automation, and executive intelligence into one deployable operating layer for lean businesses.